

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented test automation scripts using **Python**, adhering to platform standards and contributing to a **15% increase** in automated test coverage for system features.
- Executed system-level testing on automotive camera systems and mechatronic ECUs, meticulously documenting defects and reporting on execution status for assigned features.
- Collaborated with System Engineers and Function Owners to review system requirements and architecture, assessing testability and planning comprehensive test coverage.
- Developed test specifications that fulfilled system functional and performance requirements, ensuring traceability to system requirements throughout the project development lifecycle.
- Contributed to the specification of test environment requirements for **Bench** and **HIL** environments, working with Test Tool Developers to ensure readiness.
- Monitored testing progression, proving status of test completion and deviations, and maintaining test results, logs, and reports across the project lifecycle.

SKILLS

Programming Languages – C++, C, Python, C#, .NET

Testing & Validation – System Testing, Functional Requirements, Performance Requirements, Test Automation, Stress Testing, Robustness Testing, Trace Analysis, Troubleshooting

Automotive Systems – ADAS, Automotive Camera Systems, Mechatronic ECU Systems, Vehicle Environments

Development Lifecycle – Systems Development Life Cycle, Project Life Cycle, Requirements Review, Architecture Design, Test Planning

Tools & Environments – Bench Environment, HIL, NI LabVIEW, NI TestStand, Git, Jira, Confluence

PROJECTS

System Validation Framework for Automotive Software

- Developed a reusable **Python**-based framework for system-level testing of automotive software features, enabling repeatable execution across diverse hardware configurations.
- Implemented automated test case generation for functional requirements, reducing manual test design effort by approximately **20%** for new feature releases.
- Integrated the framework with **CI/CD** pipelines to provide rapid feedback on code changes, ensuring test capability and readiness for each development phase.

Performance Testing of Mechatronic Control Units

- Designed and executed performance and robustness tests for automotive mechatronic ECUs, simulating high-load conditions to identify system bottlenecks.
- Analyzed test results and provided detailed reports on detected defects and performance deviations, facilitating root cause analysis and validation of fixes.
- Maintained test environment configurations and contributed to troubleshooting efforts, ensuring the accuracy and reliability of performance testing data.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)